Bond Enthalpies

Answer key with worked solutions

A. Ideas

1. Bond enthalpy.

> The energy needed to **break one mole** of a bond in gaseous molecules to form gaseous atoms. Breaking a bond needs energy to overcome the attraction between the nuclei and the shared electrons (**endothermic**); forming a bond releases the same amount of energy (**exothermic**).

2. Bond order.

> Length, shortest first: **C≡C, C=C, C–C**. Strength, weakest first: **C–C, C=C, C≡C**. The second bond in C=C is a **π bond**, formed by side-on overlap, which is weaker than the σ bond, so C=C (614 kJ/mol) is less than 2 × 348 = 696 kJ/mol.

B. Estimating ΔH

3. $H_2 + F_2$.

> Broken: $436\ \text{kJ/mol} + 155\ \text{kJ/mol} = 591\ \text{kJ/mol}$. Formed: $2 \times 567\ \text{kJ/mol} = 1134\ \text{kJ/mol}$.
> $\Delta H \approx 591\ \text{kJ/mol} - 1134\ \text{kJ/mol} = \mathbf{-543\ kJ/mol}$

4. Chlorination of methane.

> Broken: C–H + Cl–Cl $= 413\ \text{kJ/mol} + 242\ \text{kJ/mol} = 655\ \text{kJ/mol}$. Formed: C–Cl + H–Cl $= 328\ \text{kJ/mol} + 431\ \text{kJ/mol} = 759\ \text{kJ/mol}$.
> $\Delta H \approx 655\ \text{kJ/mol} - 759\ \text{kJ/mol} = \mathbf{-104\ kJ/mol}$

5. Hydrogenation of ethyne.

> The two C–H bonds of ethyne are unchanged, so count only the bonds that change.
> Broken: C≡C + 2 H–H $= 839\ \text{kJ/mol} + 2 \times 436\ \text{kJ/mol} = 1711\ \text{kJ/mol}$
> Formed: C–C + 4 new C–H $= 348\ \text{kJ/mol} + 4 \times 413\ \text{kJ/mol} = 2000\ \text{kJ/mol}$
> $\Delta H \approx 1711\ \text{kJ/mol} - 2000\ \text{kJ/mol} = \mathbf{-289\ kJ/mol}$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

6. Combustion of ethane.

> Broken: 2 C_2H_6 contain 2 C–C and 12 C–H; plus 7 O=O:
>
> $$\begin{aligned}&(2 \times 348 + 12 \times 413 + 7 \times 498)\ \text{kJ}\\ &= (696 + 4956 + 3486)\ \text{kJ}\\ &= 9138\ \text{kJ}\end{aligned}$$
>
> Formed: 4 CO_2 contain 8 C=O; 6 H_2O contain 12 O–H:
>
> $$\begin{aligned}&(8 \times 799 + 12 \times 463)\ \text{kJ}\\ &= (6392 + 5556)\ \text{kJ}\\ &= 11\,948\ \text{kJ}\end{aligned}$$
>
> $\Delta H \approx 9138\ \text{kJ} - 11\,948\ \text{kJ} = \mathbf{-2810\ kJ}$ for the equation as written, which is **-1405 kJ per mole of ethane**.

C. Working backwards

7. H–I.

> $$\begin{aligned}-9\ \text{kJ/mol} &= (436 + 151)\ \text{kJ/mol} - 2x\\ 2x &= 587\ \text{kJ/mol} + 9\ \text{kJ/mol} = 596\ \text{kJ/mol}\\ x &= \mathbf{298\ kJ/mol}\end{aligned}$$

D. Challenge

8. Why the difference.

> (1) The measured value forms **liquid** water: condensing water vapour releases extra energy, which the gas-phase bond enthalpies leave out. (2) The bond enthalpies are **averages** over many compounds, not the exact values for the bonds in methane, carbon dioxide and water.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.